# Electronic supplementary material for "Silent but deadly? Physics says seldom, memory says mostly"

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## S1 Odorant chemistry

### S1.1 Methods

*Odour activity.* Each sulfur gas contributes an odour activity value (OAV), its mole fraction over its detection threshold, $\mathrm{OAV}_i = x_i/\theta_i$. The passage's OAV is their sum; we assume additivity and do not model mixture suppression. We compare two measured compositions:

- 87 passages of 16 volunteers [S1], used throughout the dispersion model;
- 28 passages measured by gas chromatography: $H_2S$ 149, methanethiol 216 and dimethyl sulfide 144 $\mathrm{nmol\,L^{-1}}$ [S2].

Three threshold sets bracket the observer: a trained panel [S3], population geometric means [S4, S5] and the lowest values in the literature, as compiled in ref. S2.

*Removal check.* Zinc acetate precipitates $H_2S$ and thiols; activated charcoal adsorbs all three gases. For both treatments, ref. S1 reports the gas concentrations and two judges' ratings before and after. We predict the rating ratio as $(\mathrm{OAV_{after}}/\mathrm{OAV_{before}})^n$ with $n = 0.20$–$0.45$ [S6], without fitting.

*Buoyancy.* The reduced gravity $g' = g(\rho_a - \rho_g)/\rho_a$ of each of 16 measured compositions [S7] is split into a thermal part, $g(1-T_a/T_g)$, for gas at 37 °C in air at 22 °C, and a compositional part, $g(M_a - M_g)/M_a$, at equal temperature with air at 50% relative humidity. The water-vapour content of flatus has not been measured, so we bracket dry and saturated gas.

*Gas-phase and surface losses.* Gas-phase losses use IUPAC rate constants at 298 K for reaction with OH, $O_3$, $NO_3$ and Cl [S8], with oxidant levels measured in an office: OH 0.6–$9.8 \times 10^5\,\mathrm{cm^{-3}}$ and $O_3$ 0.7–2.6 ppb [S9]. Two assumed upper cases add OH $10^6\,\mathrm{cm^{-3}}$ with $O_3$ 40 ppb, and Cl $10^6\,\mathrm{cm^{-3}}$ during hypochlorite cleaning. No deposition velocity $v_d$ has been published for these gases on indoor surfaces, so surface loss is bracketed as $k = v_d\,S/V$ with $S/V = 3\,\mathrm{m^{-1}}$.

*Dilution in a panel test.* Ref. S2 reports the lowest source concentration that every panel member detected when 100 mL was released below the nose. Divided by a detection threshold, it gives the dilution between release and nose in that test, which we compare with the breath-averaged dilution of the analytic model.

### S1.2 Results

*Which molecule.* Over the two measured compositions and three threshold sets, methanethiol carries 43–94% of the summed OAV and never less than $H_2S$ (Figure S1a).

- With the composition of ref. S1, methanethiol and $H_2S$ carry about equal shares (43–54% and 43–46% with trained-panel and population thresholds). Methanethiol is one fifth as concentrated, but its threshold is 5–6 times lower.

- With the composition of ref. S2, or with the lowest reported thresholds, methanethiol carries 60–94%.

The two datasets disagree on the concentrations themselves: $H_2S$ is five times methanethiol in one and slightly below it in the other, a difference ref. S2 attributes to the bean-rich diet of the first. In that study's panel test, gas mixtures with methanethiol above its flatus threshold smelled like most passages.

Composition does not enter the silent/audible comparison: transport is linear and acts on all three gases alike, so every ratio is the same for any composition. Absolute OAVs are 11–38% lower with the second composition.

*The intensity law with real flatus.* Zinc acetate lowers the summed OAV by a factor of 14. Stevens' law then predicts rating ratios of 0.31–0.59, and the two judges gave 0.28 and 0.59 [S1]. Activated charcoal lowers the OAV about 260-fold and is rated far weaker than the power law predicts, as power laws fail near threshold. The silent/audible comparison uses supra-threshold ratios, where the law held.

*What makes a passage rise or sink.* Composition sets most of the buoyancy (Figure S1b).

- For the median passage the compositional part of the reduced gravity ($+1.25\,\mathrm{m\,s^{-2}}$ dry, +1.40 saturated) is about three times the thermal part (+0.47).
- $H_2$ and $CH_4$ make flatus light and $CO_2$ makes it heavy; the compositional part falls with $CO_2$ content ($r = -0.80$).
- 6–12% of passages sink even while hot. After cooling to room temperature, 25–38% are heavier than air.

The dispersion model uses these signed values of $g'$.

*Losses on the way to the nose* (Table S1).

- In an office, gas-phase oxidation removes at most 0.4% of any of the three gases in 120 s, and hypochlorite cleaning less than 4%.
- Surface deposition is unconstrained by data. It removes 0.7% in 120 s at $v_d = 0.002\,\mathrm{cm\,s^{-1}}$ and 51% at $0.2\,\mathrm{cm\,s^{-1}}$. Treating the odorants as passive scalars therefore holds against chemistry by two to three orders of magnitude, and against deposition only if $v_d \lesssim 0.01\,\mathrm{cm\,s^{-1}}$.
- Deposition would remove most from the silent release, which stays close to the body and the seat, and would lower its doses further in the simulated scenes.

*How dilute a passage arrives.* The panel test of ref. S2 implies a dilution of $0.6\text{–}13 \times 10^4$ between release and nose, depending on the threshold set. The breath-averaged dilution of the analytic model at the neighbour has a median of $1.8 \times 10^3$ (10–90%: $2.5 \times 10^2$–$2.9 \times 10^4$). The panel value is biased upwards, because it requires the least sensitive panellist to detect but divides by the thresholds of the most sensitive. The analytic model is at the concentrated end of the panel range, as intended for an upper bound, and within about an order of magnitude of it.

*Composition and volume.* Larger passages contain more fermentation gas and less nitrogen (per passage, $r = +0.56$ for $CO_2$ and $-0.50$ for $N_2$, $n = 87$), while the sulfur concentrations do not change with volume ($|r| \leq 0.22$, not significant); sulfur mass therefore rises with passage volume [S7]. At high production rates almost all flatus is fermentation gas [S10]. The fermentation-gas fraction rises with output under bean, fibre and lactulose challenges in every study that measured both ($r = +0.86$ to $+0.99$, computed from the published tables [S11–S16]; in one person from 26% at 55 mL to 79% at 720 mL per 4 h [S17]), whereas on an ordinary diet larger outputs carry more nitrogen [S18]. The density of the gas does not follow the amount:

Table S1: Gas-phase lifetime of the odorants and the fraction lost in 120 s. Office oxidant levels are measured [S9]: low, OH $0.6 \times 10^5$ cm$^{-3}$ and $O_3$ 0.7 ppb; high, OH $9.8 \times 10^5$ cm$^{-3}$ and $O_3$ 2.6 ppb. The cleaning case assumes Cl $10^6$ cm$^{-3}$. Rate constants from ref. S8.

| scenario | $H_2S$ | $CH_3SH$ | DMS |
|---|---|---|---|
| office, low | 40 d / 0.003% | 5.7 d / 0.03% | 28 d / 0.005% |
| office, high | 2.5 d / 0.06% | 8.6 h / 0.4% | 1.8 d / 0.08% |
| hypochlorite cleaning | 3.5 h / 0.9% | 1.2 h / 2.8% | 0.8 h / 4.0% |

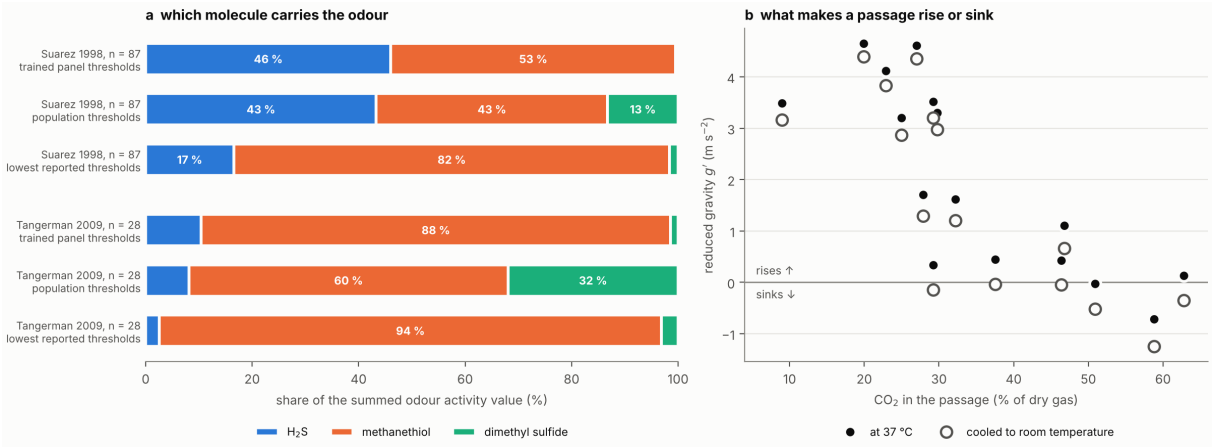



Figure S1: Odorants and buoyancy. (a) Share of the summed OAV carried by $H_2S$, methanethiol and dimethyl sulfide, for two measured compositions [S1, S2] under three threshold sets. (b) Reduced gravity of 16 measured compositions [S7] against their $CO_2$ content, at 37 °C (dry basis; water saturation shifts these values by $-0.04$ to $+0.29$ m s$^{-2}$) and after cooling to room temperature.

carbon dioxide and hydrogen rise together, and the mean molar mass, 22–32 g mol$^{-1}$ across diets, is set by the substrate's $CO_2$:$H_2$ ratio (its correlation with output ranges from $-0.37$ to $+0.90$ across seven studies). Composition therefore cannot link audibility to the amount of gas, since it enters audibility only through density (Section S2).

## S2 Acoustics

*Duration and orifice.* Flatus passes during an anal relaxation lasting $19 \pm 6$ s [S19, S20], which bounds the duration $T$ from above, and the mid anal canal widens to about 8 mm only at balloon pressures above those that drive flatus [S21–S23], which bounds the effective diameter $d$.

*Valve.* The anal canal is modelled as a lip reed, a soft-tissue valve that self-oscillates in the flow through it, as the lips do on a brass instrument. It oscillates when the driving pressure $\frac{1}{2}\rho(u_0/C_d)^2$ exceeds a threshold $P_{th} = 0.6$–$2.5$ kPa (nominal 1.2 kPa), with discharge coefficient $C_d = 0.85$. The closest measured soft-tissue sphincter source, the pharyngo-oesophageal segment of oesophageal speech, sustains phonation down to 1.1–2.5 kPa [S24–S26], which bounds the onset from above; lips on a brass instrument start at about 0.6 kPa [S27]. This gives an onset velocity $u_{onset} \approx 42$ m s$^{-1}$ (21–71).

An oscillating valve radiates as a harmonic monopole with harmonics falling by 12 dB per octave. Its fundamental is taken as 160 Hz (60–300 Hz), between oesophageal speech at 60–150 Hz [S24, S28–S30] and recordings of flatus near 270 Hz, which are not peer-reviewed. A release counts as *audible* when at least one harmonic exceeds both the ISO 226 hearing threshold [S31] and the masked threshold of the background, computed from the critical ratio [S32].

Backgrounds range from a 35 dB(A) living room to a 77 dB(A) subway car.

*Valve state.* Manometry shows flatus passing through a relaxed sphincter at a rectal pressure rise of 0–15 mmHg [S19, S23] but also against a contracted one [S33]. The 85% silent share treats the onset pressure as a property of the tissue and does not distinguish the two.

## S3 CFD validation

- *Free round jet* (axisymmetric, Re = 6700, three meshes). The velocity-decay constant is 5.0 and the scalar constant $K_c = 4.3$–$4.6$, with a grid uncertainty of at most 3.7%. The $k$–$\omega$ SST model spreads round jets about 25% too fast, which over-dilutes audible releases by about 15% and so favours the silent release.

- *Convective boundary layer* of a simplified seated manikin with skin at 33 °C, compared with thermal-manikin data [S34]. The face-front peak velocity is 40–56% too high at room temperatures of 20–26 °C; its change with room temperature is reproduced ($-27$% against $-33$% measured). The fast boundary layer shortens travel times to the releaser's own nose by about 30%, and the analytic model stretches the CFD travel times by a factor of 1.4 to correct for it.

## S4 CFD scenes

All scenes use the solver of the main text, skin at 33 °C and walls at 23 °C. Each is first run without a release until the body plumes have developed (30 s; 90 s with ventilation). From that flow field the same 92 mL is released behind the releaser's buttocks, pointing down and back, as an inlet that matches the release's volume flow and momentum: silent, 2 s through 8 mm ($u_0 = 0.9\,\mathrm{m\,s^{-1}}$); audible, 0.5 s through 2.4 mm ($41\,\mathrm{m\,s^{-1}}$). The odorant is a passive scalar, sampled 2.5 cm in front of each nose, over 5 cm discs at nose height (breathing zone) and over sinusoidal breathing cycles; the worst breath is the most concentrated single inhalation.

- *Sofa.* A $3.2 \times 3.2 \times 2.6$ m living room with a 0.5 m ceiling vent for pressure relief (size and position assumed). Two seated manikins sit 0.6 m apart on a two-seat sofa whose base is solid to the floor. Mesh 0.72 million cells, refined to 5 mm at the release and 2.5 mm on the inlet face. Runs of 30 s, the upright one extended to 120 s. Two variants rotate the releaser's upper body about the hip: reclined by 4.5 °, so that the back closes onto the backrest, and leaning forward by 10 °, so that the gap behind the back opens.

- *Elevator.* The car of a 630 kg, eight-person elevator, 1.10 m wide, 1.40 m deep and 2.10 m high (a manufacturer's standard car, within the maximum area of 1.66 $\mathrm{m^2}$ allowed for that load). Four standing manikins (nose at 1.59 m) stand in a $2 \times 2$ grid facing the door; the fan is off, and a 0.2 m ceiling opening relieves pressure (size assumed). Mesh 0.91 million cells. Runs of 60 s with the release behind a front-row or a back-row occupant; mirror symmetry gives the other two positions.

- *Standing pair.* The sofa room without the sofa, with two standing manikins face to face, noses 1.0 m apart and 0.8 m of open floor behind each back. Still air and office ventilation share one mesh of 0.43 million cells. Ventilation is one square ceiling diffuser above the pair supplying $21.2\,\mathrm{L\,s^{-1}}$, i.e. 2.9 air changes per hour (EN 16798-1 Category II for two people in a low-polluting 10 $\mathrm{m^2}$ room: $7\,\mathrm{L\,s^{-1}}$ per person plus $0.7\,\mathrm{L\,s^{-1}\,m^{-2}}$ [S35]). The air leaves the diffuser along the ceiling at an assumed $1.5\,\mathrm{m\,s^{-1}}$, at room temperature, and exits through a ceiling grille. Runs of 60 s.

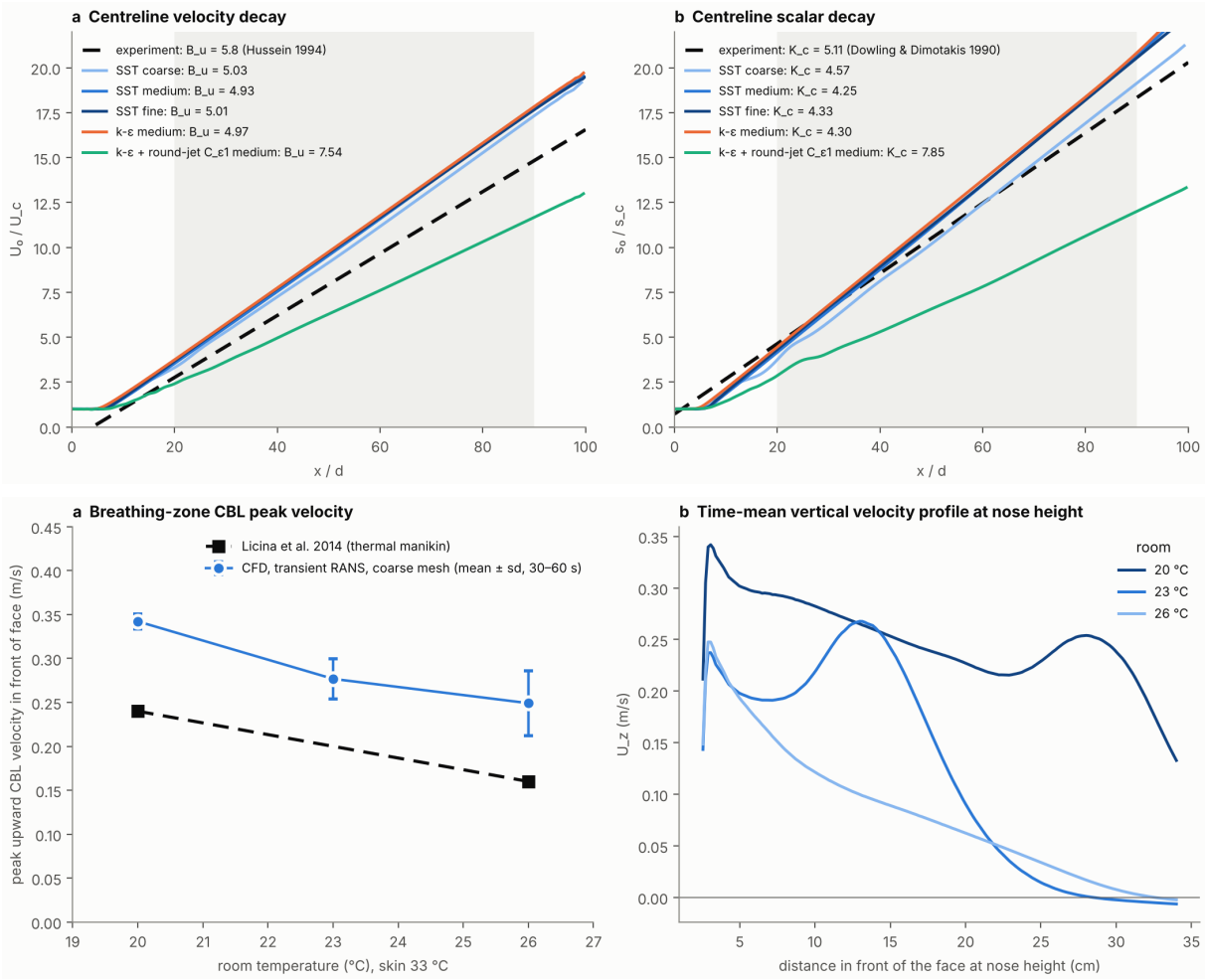



Figure S2: Validation. Top: free round jet at Re = 6700 on three meshes (centreline velocity and scalar decay, spreading). Bottom: face-front convective boundary layer of the seated manikin against thermal-manikin data [S34] at 20–26 °C, and time-mean vertical velocity profiles at nose height.

Table S2: Sofa: peak breathing-zone OAV within 30 s of the release (mean flow).

| nose | posture | silent | audible |
|---|---|---|---|
| releaser | upright | $5 \times 10^{-11}$ | 0.46 |
| releaser | reclined 4.5 ° | $9 \times 10^{-11}$ | 3.3 |
| releaser | forward 10 ° | $3 \times 10^{-7}$ | $1.5 \times 10^{-5}$ |
| neighbour | upright | $3 \times 10^{-14}$ | 0.63 |
| neighbour | reclined 4.5 ° | $3 \times 10^{-15}$ | 0.19 |
| neighbour | forward 10 ° | $1 \times 10^{-4}$ | 2.6 |

## S5 Sofa

*Silent release.* The silent release stays in the convective boundary layer of the releaser's back. It rises in the gap between back and backrest, over the head and to the ceiling. Within 30 s its breathing-zone OAV is at most $10^{-10}$ at both noses, while within 25 cm of the releaser's head it reaches 6 000.

*Audible release.* The audible jet hits the corner between seat and backrest, spreads sideways along the gap and rises up both backs as a wider, more dilute sheet; part of it curls around the heads. It reaches the neighbour's head region within 6 s and both breathing zones at OAV 0.2–0.6.

*Reclined by 4.5 °.* With the back closed onto the backrest, the silent release leaves the seat-level wedge sideways and still rises behind the head; no breathing zone receives more than $10^{-10}$. The audible jet wraps around the releaser's face and exceeds threshold at their own nose (OAV 2–4).

*Leaning forward by 10 °.* With the gap behind the back open, both releases rise behind the head, and the releaser's nose, 13 cm further forward, receives neither (breathing-zone OAV $\leq 10^{-5}$). The silent cloud reaches the region within 25 cm of the neighbour's head as strongly as the audible one (OAV 580 against 450) but not the neighbour's breathing zone ($10^{-4}$); the audible jet reaches it at OAV 2.6.

*120 s.* Both clouds reach the ceiling, return through the room and arrive at both noses from about 45 s. After about 70 s the silent and audible traces at the neighbour are indistinguishable (worst breath 0.39 for both). At the releaser's own nose the mixed silent cloud is 1.5–2 times higher and touches the trained-panel threshold for about a second near 100 s. These levels are 10–23 times below the thresholds of untrained observers.

## S6 Elevator

A silent release follows the releaser's own back plume, so what stands behind the releaser decides where it goes.

- With a person behind (front-row releaser), it rises in the 0.2 m gap between the releaser's back and that person's face, into their breathing zone. The silent release is stronger at every nose ($R = 1.4$–$4.3$), most of all for the person behind (OAV 27).
- With the car wall behind (back-row releaser), it rises between back and wall and reaches the front row only below threshold. The audible jet hits the floor, is lifted by all four body plumes, and is stronger at every nose ($R = 0.20$–$0.55$).

With a front-row releaser the person behind reaches the trained-panel threshold first (8.8 s), before the releaser (17.6 s); with a back-row releaser the releaser is first. Whoever smells it first is therefore the releaser in half of the positions, for both modes.

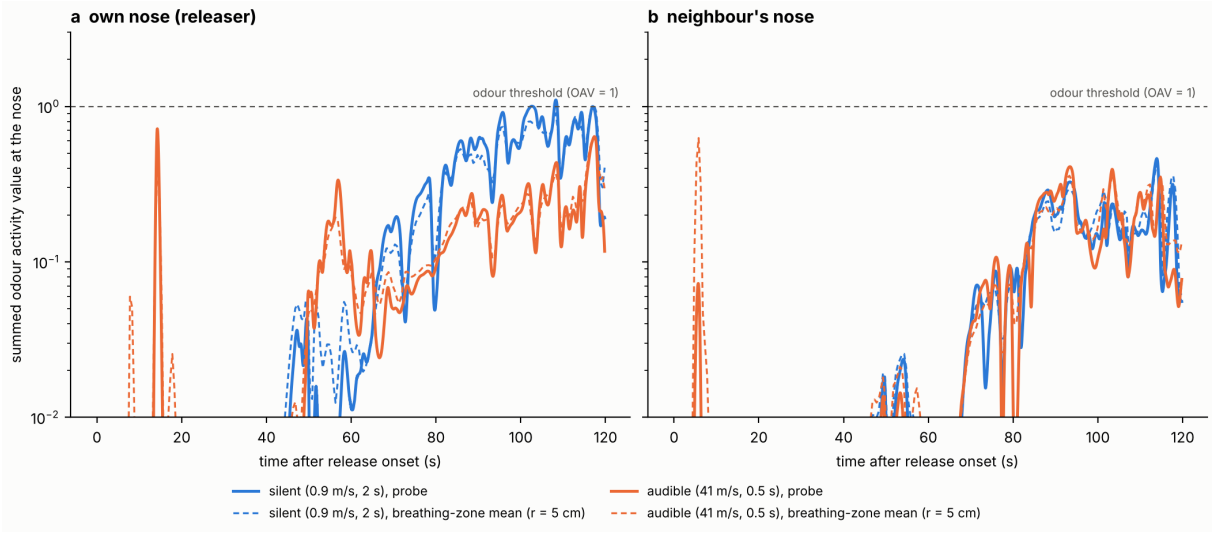



Figure S3: Sofa, upright: OAV at the releaser's and the neighbour's nose over 120 s for the same 92 mL released silently or audibly. Solid: 2.5 cm in front of the nose; dashed: mean over a 5 cm breathing-zone disc. The horizontal line is the trained-panel odour threshold.

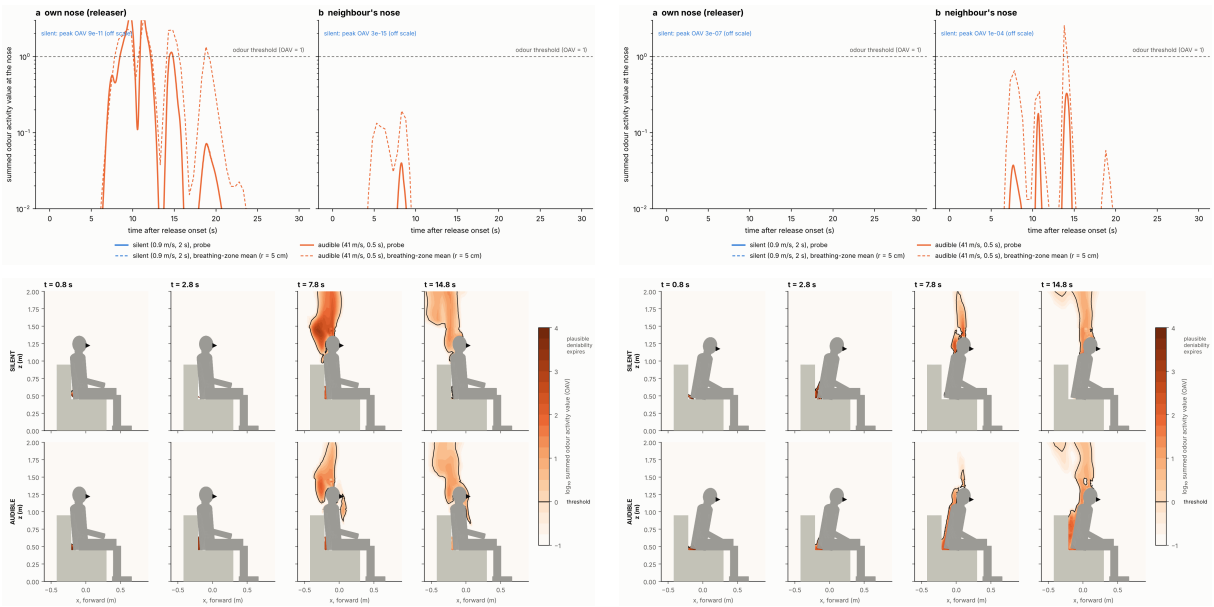



Figure S4: Sofa, releaser reclined by 4.5 ° (left) or leaning forward by 10 ° (right), 30 s. Top: OAV at both noses, as in Figure S3. Bottom: $\log_{10}$ OAV in the releaser's mid-sagittal plane, silent (upper rows) and audible (lower rows).

Table S3: Elevator: expected worst-breath OAV within 60 s at each nose (trained-panel thresholds; mean flow), for a front-row and a back-row releaser.

| releaser | nose | silent | audible | $R$ |
|---|---|---|---|---|
| front row | releaser | 6.7 | 1.9 | 3.4 |
| | beside | 3.6 | 0.8 | 4.3 |
| | behind | 27 | 20 | 1.4 |
| | diagonal | 7.6 | 2.7 | 2.9 |
| back row | releaser | 4.5 | 23 | 0.20 |
| | beside | 2.3 | 5.9 | 0.39 |
| | in front | 0.57 | 2.8 | 0.20 |
| | diagonal | 0.41 | 0.74 | 0.55 |

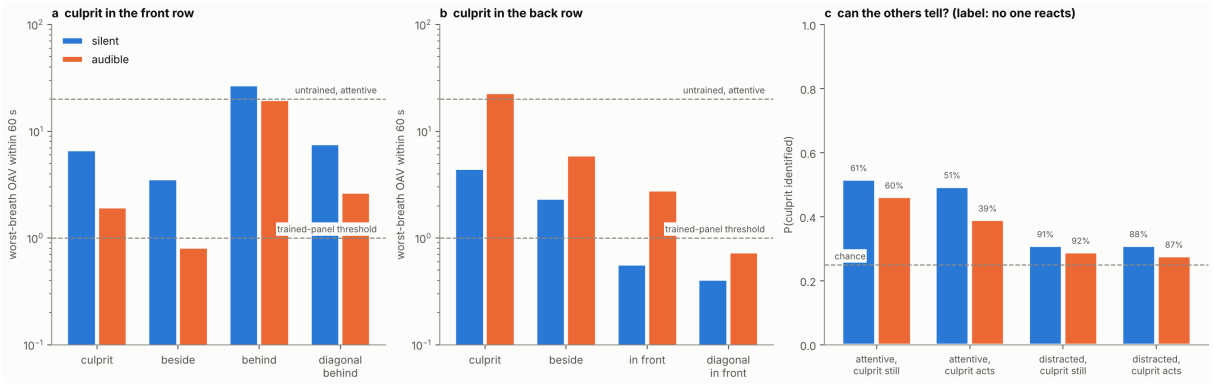



Figure S5: Elevator: four standing occupants in a closed 630 kg car, fan off, same 92 mL. (a, b) Expected worst-breath OAV within 60 s at each nose for a front-row and a back-row releaser; dashed lines mark the trained-panel threshold and the attentive population threshold. (c) Probability that the others identify the releaser from who reacted, for attentive and distracted onlookers and a releaser who stays still or reacts along; labels give the probability that no one reacts. The dashed line is chance.

*Identifying the releaser.* Each occupant other than the releaser reacts with the detection probability of the main text's observer models, independently. Onlookers see only who reacted and infer the releaser with a uniform prior, for a releaser who never reacts and one who reacts like the others. With distracted occupants no one reacts in 87–92% of passages, and the releaser is identified at 0.28–0.31; attentive occupants identify the releaser at 0.39–0.52 (chance 0.25). Silent and audible passages are identified about equally well by odour; the sound of an audible passage, which also points to the releaser, is not modelled.

## S7 Standing pair

Results pending (simulations running).

## S8 Breath sampling

An audible puff reaches the neighbour after a median of 16 s and passes with a time scale of about 3.9 s (10–90%: 1.5–12 s). Without a warning, the breathing phase barely matters: for a median passage the worst breath captures 81–99% of the peak. A listener warned by the sound needs about 25 s of breath-hold for an even chance of missing the puff (worst breath below 10% of the peak); 33 s, the typical maximum at end-expiration, gives 62%.

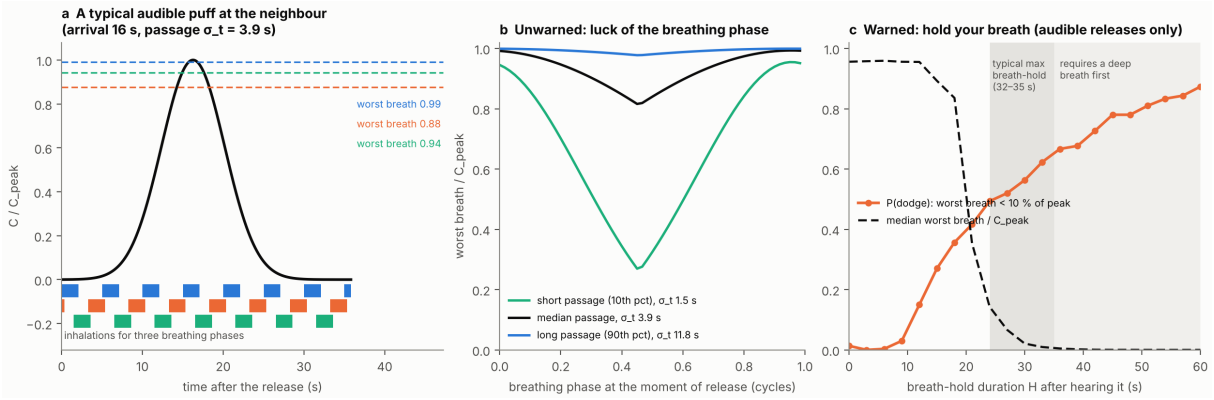



Figure S6: Breath sampling at the neighbour's nose for audible releases. (a) A typical puff (arrival 16 s, passage time scale 3.9 s) and the inhalations of three breathing phases, with the worst breath of each. (b) Without warning: worst breath relative to the peak against the breathing phase at release, for short, median and long passages. (c) Warned: probability of missing the puff (worst breath below 10% of the peak) and median worst breath, against the breath-hold duration after hearing the release (0.5 s reaction latency).

## S9 Origin of the odorants

Two facts the transport model takes as given have chemical explanations in the literature; we did not test them.

*Why sulfur concentration does not follow volume.*

- $H_2S$ comes from sulfate-reducing bacteria and from cysteine degradation, whose genes are widespread in the gut microbiome [S36, S37]. Sulfate reducers and methanogens both consume $H_2$, but whether they exclude each other is disputed. Sulfidogenic gut contents produced no methane in some studies [S36, S38], whereas faecal sulfide did not differ between methane producers and non-producers in others [S39–S41], and methanogenic faeces suppressed sulfide production sixfold in co-incubation [S42].

- The caecal mucosa of rats removes both gases quickly: methanethiol is demethylated to $H_2S$, and $H_2S$ is oxidised mainly to thiosulfate [S43]. Rhodanese is the main detoxifying enzyme in human colon [S44], and SELENBP1 oxidises methanethiol with an apparent $K_m$ of 4.8 nM [S45].

- Only free sulfide can enter the gas, and it is less than 5% of faecal total sulfide [S2].

The odorants are thus trace products held at a balance of production and fast consumption, whereas passage volume is set by $N_2$, $CO_2$, $H_2$ and $CH_4$. We infer that this is why sulfur concentration and passage volume are uncorrelated [S7]. Diet moves the balance: a meat-rich diet raised faecal sulfide from 0.22 to 3.38 mmol kg$^{-1}$ in five volunteers [S46], although faecal sulfide reflects the binding capacity of the stool as much as sulfide production [S47], and the response to animal and plant diets differs between people [S48].

*Diet can make large passages less sulfurous.* Within one diet, sulfur concentration does not change with passage volume (Section S1); across diets it can. Rapidly fermented fibres raised gas production up to 50-fold in faecal incubations while suppressing $H_2S$ by up to 89%, and cysteine raised $H_2S$ 16-fold [S49]. Dietary sulfate tripled sulfide production and cut methane production fivefold, at unchanged stool weight and transit, in three of six subjects [S50]; in sulfidogenic gut contents $H_2S$ replaced $CH_4$ almost mole for mole [S38], diverting hydrogen from gas volume to sulfide [S51]. Interventions aimed at sulfur move sulfur alone: bismuth cut faecal $H_2S$ release 23-fold with total gas unchanged [S52], and in dogs charcoal, yucca and zinc acetate cut $H_2S$

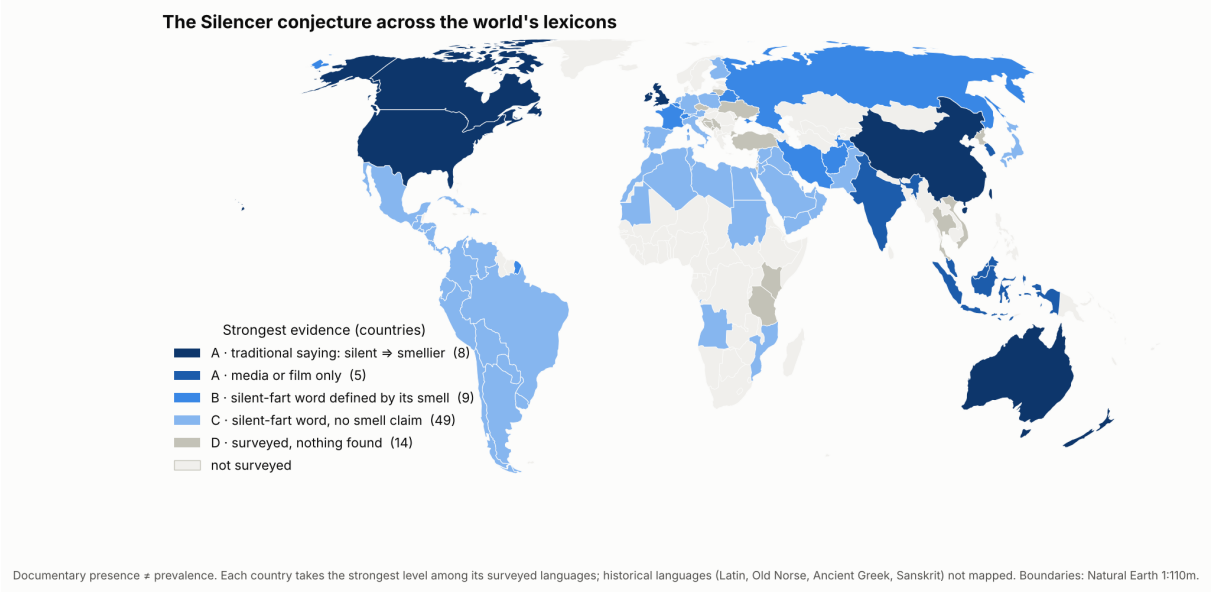



Figure S7: The belief across languages, by evidence level. Each country takes the strongest level among its surveyed languages. A: an explicit saying that silent passages smell worse (traditional, or in media or film only); B: a dedicated word for a silent passage defined by its smell; C: a dedicated word with no smell claim; D: surveyed, nothing found.

by 38–86% without changing gas production or the number of emissions [S53]. Because audible passages are larger, a diet-driven link of this kind would make audible passages less sulfurous per millilitre, in the direction of the belief. Its size in vivo is unknown: no study has measured sulfur gases and flatus volume together across diets.

*Why methanethiol smells foul.* The human receptor OR2T11 responds to short thiols, including methanethiol, in a copper-dependent way, but not to $H_2S$ [S54]. Human nasal mucus contains about 2.5 µM copper [S55], below the 30 µM added in the receptor assays, so the receptor's role in vivo is plausible but not shown. Other receptors respond to sulfur ligands without copper [S56–S58]. A receptor that responds to methanethiol and not to $H_2S$ would make methanethiol the characteristic note of flatus independently of thresholds; we infer this, it has not been tested.

## S10 Cross-cultural survey

We searched dictionaries, proverb collections and etymological handbooks in 32 languages or historical stages for a dedicated word for a silent passage and for an explicit claim that it smells worse, and checked every entry used against a retrieved source. Each language is graded by its strongest evidence: A, an explicit saying (traditional, or in media or film only); B, a dedicated word defined by its smell; C, a dedicated word without a smell claim; D, nothing found. The silent/loud distinction is lexicalised in at least five language families. The explicit claim is traditional only in Chinese and English, and smell-defined silent words arose independently at least four times within Indo-European. The survey records whether the belief is documented in a language, not how many speakers hold it.

## S11 Code and data

Code, inputs and processed outputs are in the repository named in the Data accessibility statement of the main text; its `README.md` lists the source files of every result. They are not duplicated here.

## S12 Parameters

Table S4: Model parameters: nominal value, range sampled or tested, and basis: taken from a cited measurement, derived from cited data, partly sourced, taken from a non-peer-reviewed source, or assumed. Parameter names are those of the code; the source and notes for every entry are in `params/params.yaml` of the code archive.

| parameter | nominal | range | units | basis |
|---|---|---|---|---|
| *Source* | | | | |
| `T_gas` | 310.15 | 309.85–310.65 | K | measured |
| `V_passage` | $9.2 \times 10^{-5}$ | $1.7 \times 10^{-5}$–$3.75 \times 10^{-4}$ | $m^3$ | measured |
| `V_daily` | $7.05 \times 10^{-4}$ | $4.76 \times 10^{-4}$–0.001491 | $m^3\,d^{-1}$ | measured |
| `n_daily` | 10 | 1–20 | $d^{-1}$ | measured |
| `d_orifice` | 0.004 | 0.001–0.008 | m | partly sourced |
| `T_release` | 1 | 0.3–20 | s | partly sourced |
| `dp_rectal` | 2000 | 500–4000 | Pa | assumed |
| `p_anal_rest` | 8666 | 6133–12000 | Pa | measured |
| `g_reduced` | 1.7 | -0.66–4.68 | $m\,s^{-2}$ | derived |
| `nu_gas` | $1.7 \times 10^{-5}$ | | $m^2\,s^{-1}$ | assumed |
| `c_H2S` | 1.06 | 0.26–1.86 | $\mu mol\,L^{-1}$ | measured |
| `c_CH3SH` | 0.21 | 0.05–0.37 | $\mu mol\,L^{-1}$ | measured |
| `c_DMS` | 0.08 | 0.04–0.12 | $\mu mol\,L^{-1}$ | measured |
| `sulfur_sigma_ln` | 1.1 | 1–1.25 | – | derived |
| *Odour thresholds* | | | | |
| `H2S` | 0.41 | 0.41–0.9 | ppbv | measured |
| `CH3SH` | 0.07 | | ppbv | measured |
| `DMS` | 3 | | ppbv | measured |
| *Jet and puff* | | | | |
| `B_u` | 5.8 | 5.8–5.9 | – | measured |
| `S_jet` | 0.094 | 0.094–0.102 | – | measured |
| `K_c` | 5.11 | 4.73–5.2 | – | measured |
| `Gamma_sj` | 2.8 | 2.74–2.86 | – | measured |
| `C_puff` | 2.6 | 2.54–2.66 | – | measured |
| `alpha_th` | 0.25 | 0.15–0.35 | – | measured |
| `alpha_jet` | 0.07 | 0.065–0.08 | – | measured |
| `alpha_plume` | 0.12 | 0.1–0.16 | – | measured |
| *Environment* | | | | |
| `u_handoff` | 0.15 | 0.05–0.25 | $m\,s^{-1}$ | assumed |
| `u_CBL` | 0.17 | 0.11–0.24 | $m\,s^{-1}$ | measured |
| `u_CBL_pelvis` | 0.1 | 0.05–0.15 | $m\,s^{-1}$ | assumed |
| `u_own_eff` | 0.11 | 0.07–0.16 | $m\,s^{-1}$ | derived |
| `delta_CBL` | 0.1 | 0.05–0.15 | m | assumed |
| `u_room` | 0.08 | 0.03–0.15 | $m\,s^{-1}$ | assumed |
| `K_turb` | 0.002 | $1 \times 10^{-4}$–0.013 | $m^2\,s^{-1}$ | measured |
| `D_H2S_air` | $1.65 \times 10^{-5}$ | $1.62 \times 10^{-5}$–$1.77 \times 10^{-5}$ | $m^2\,s^{-1}$ | partly sourced |
| *Geometry* | | | | |
| `eye_sit_m` | 0.804 | 0.75–0.86 | m | measured |
| `eye_sit_f` | 0.748 | 0.698–0.799 | m | measured |
| `eye_stand_m` | 1.64 | 1.536–1.753 | m | derived |
| `eye_stand_f` | 1.517 | 1.421–1.626 | m | derived |
| `L_own` | 0.74 | 0.65–0.85 | m | assumed |
| `L_neighbour` | 1 | 0.8–1.4 | m | assumed |
| *Breathing* | | | | |
| `V_tidal` | $6.3 \times 10^{-4}$ | $4 \times 10^{-4}$–$8.3 \times 10^{-4}$ | $m^3$ | measured |
| `f_breath` | 12 | 10–16 | $min^{-1}$ | measured |

Table S4: Model parameters (continued).

| parameter | nominal | range | units | basis |
|---|---|---|---|---|
| `Ti_Ttot` | 0.42 | 0.34–0.5 | – | measured |
| `T_insp` | 2.1 | 1.6–2.7 | s | derived |
| *Acoustics* | | | | |
| `f0_flatus` | 160 | 60–300 | Hz | partly sourced |
| `bg_library` | 40 | 35–45 | dB(A) | assumed |
| `bg_living` | 35 | 30–45 | dB(A) | assumed |
| `bg_office` | 45 | 43–55 | dB(A) | measured |
| `bg_restaurant` | 74 | 67–83 | dB(A) | measured |
| `bg_subway` | 77 | 74–80 | dB(A) | measured |
| `P_th` | 1200 | 600–2500 | Pa | partly sourced |
| `C_d` | 0.85 | 0.6–1 | – | partly sourced |
| `alpha_mod` | 0.7 | 0.5–1 | – | assumed |
| `harm_slope` | -12 | -18—6 | dB/octave | assumed |
| `n_harm` | 8 | | – | assumed |
| `A_cloth` | 1 | 0–3 | dB | partly sourced |
| `r_listener` | 0.6 | 0.4–3 | m | assumed |
| *Perception and observers* | | | | |
| `n_stevens` | 0.28 | 0.2–0.45 | – | measured |
| `k_w_H2S` | 0.95 | 0.8–1.2 | intensity units per decade | partly sourced |
| `weber_C` | 0.28 | 0.05–0.4 | $\Delta C/C$ | measured |
| `tau_adapt` | 80 | 40–120 | s | assumed |
| `a_inf` | 0.35 | 0.25–0.4 | – | partly sourced |
| `sniff_ratio` | 0.64 | 0.61–0.75 | – | measured |
| `t_react` | 0.5 | 0.3–1.5 | s | assumed |
| `H_max` | 33 | 24–90 | s | measured |
| `thr_scale_typ` | 0.05 | 0.05–0.1 | – | measured |
| `thr_sd_bs` | 2 | | binary steps | measured |
| `distracted_factor` | 26 | | – | measured |
| `distracted_sd_bs` | 3.5 | | binary steps | measured |
| `recovery_gain` | 1.5 | 1–2 | – | assumed |

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